INTRODUCTION TO SYNTAX I

Session 4: Clause structure I

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TODAY'S PLAN

- Building sentences
- Functional categories
- Revising our (kinda but not quite yet clausal) structure
- Revisiting c-command

FROM PHRASES TO SENTENCES

What we know so far:

- Sentences consist of **constituents** which may be **embedded** within each other;
- Constituents bear/assign **θ-roles** + stand in a **selectional relation** with each other;
- Constituents have **heads** from which morphosyntactic features are **projected**;
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What we will learn next:

- Sentences have a core consisting of the projections of a **lexical category** $\Rightarrow$ the verb phrase (VP).
- On top of this core, **functional categories** are built that do project, but do not assign θ-roles.

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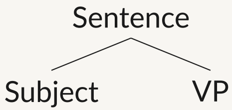

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Sentence

Subject VP

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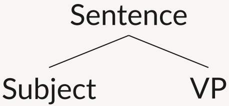

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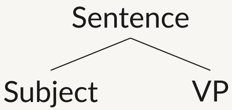

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- (1) would require us to make up a new rule about how the category *Sentence* is introduced, what its properties are, and why building it requires a different approach than combining heads, complements and specifiers.

FROM PHRASES TO SENTENCES

Recall our VP from last session:

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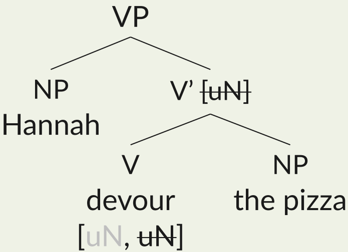

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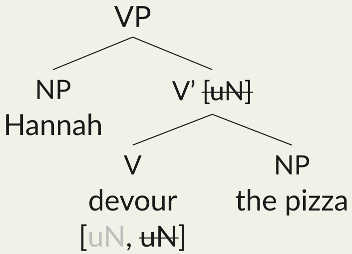

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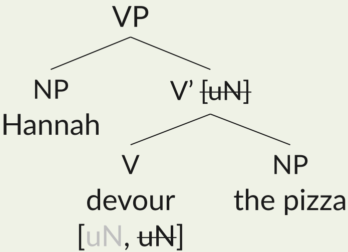

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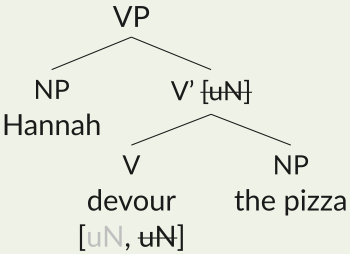

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- **Hint:** we talked an awful lot about c-selectional features. What **other morphosyntactic** features are there? (recall session 2!)
- We have yet to introduce **verbal inflection**!

LOCATING THE SOURCE OF INFLECTION

- **Verbal inflection is a property of verbs. Why do we need another head for that?**

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  - Some languages have **auxiliary verbs**: verbs with restricted semantic functions.
  - **Modal auxiliaries** express obligation, possibility, permission, intension, etc.

(3) Gilgamesh **may/must/can/should/will** seek Ishtar.

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Modals are not part of the verbal complex formed by V and its complement:

(4) a. ...and [seek Ishtar], Gilgamesh may
b. *...and [may seek Ishtar], Gilgamesh does
c. What Gilgamesh may do is [seek Ishtar][1]
d. *What Gilgamesh does is [may seek Ishtar]

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In English, expressing tense outside of the VP is optional for emphasis:

(5) a. Enkidu **did** free animals.
b. Enkidu free**d** animals.

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(6) a. Gilgamesh [$_{VP}$loved Ishtar] and Enkidu **did** [$_{VP}$], too.
    b. The mouse [$_{VP}$fears the cat] and the dog **does** [$_{VP}$], too.

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Infinitival clauses

- Although English verbal inflection is morphologically rather poor, we nevertheless distinguish between **finite** and **infinitival** sentences:

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- In the (b)-sentences, there is a ***to*** that looks like a preposition but precedes verbs[3] ⇒ it is **incompatible with** verbs that **agree or carry tense marking**.

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Summary of evidence

- Modals, emphatic *do* and *to* in infinitival clauses seem to sit **outside of the verbal complex but below the subject**.
- Morphological tense marking appears in/depends on this position.

CROSS-LINGUISTIC VARIABILITY

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In Hungarian, modality shows up on the verb when it is (un)marked for present tense, but has to escape to an auxiliary-like expletive stem when the verb is marked for past tense (12).

(12) *Hungarian*

a. Jani el-men-**ne** a bolt-ba.
   Jani PV-go-MOD DET shop-ILL
   'Jani would go to the store.'

b. Jani el-men-**t** vol-**na** a bolt-ba.
   Jani PV-go-PST EXPL-MOD DET shop-ILL
   'Jani would have gone to the store.'

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⇒ All of them of the same category [T] given their complementary distribution.

But wait... where is the subject?

- So far, we assumed that the subject is Merged in the specifier of V (recall (2)).
- Now, we saw that there seems to be a head, which we labeled T, intervening between the subject and the verbal complex.
- We keep the assumption that the subject is **base generated** in the specifier of V.

How does the subject end up above the T head?

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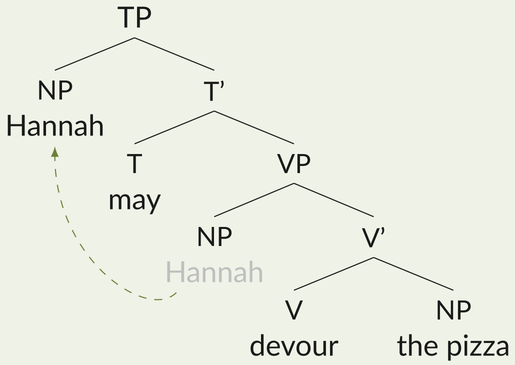

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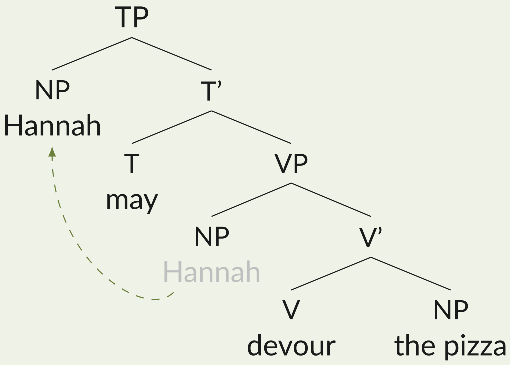

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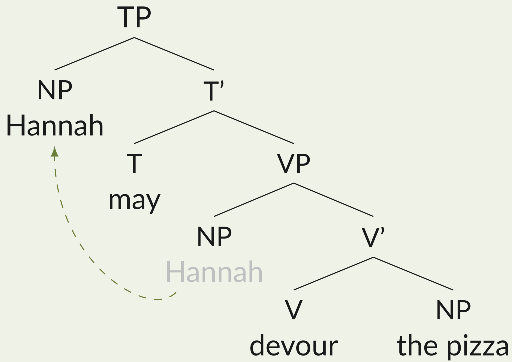

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- This structure can account for *Hannah may devour the pizza*, where *devour the pizza* is the complement of the T head *may*, linearized to the right of it; and *Hannah* is the specifier of T linearized to the left of the intermediate projection T'.

PROPERTIES OF T

T is a functional category.

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T is a functional category.

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- Although the verb *rain* does not select an argument, T requires a constituent in its specifier — hence the expletive *it*.

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(15) a. T[present] ...V[present]
b. T[past] ...V[past]
c. *T[present] ...V[past]
d. *T[past] ...V[present]

RECAP: C-COMMAND

We won't get into the formal relation between V and T.

- But: remember c-command? It is one of the prerequisites for this relation to hold.

$\Rightarrow$ Node α **c-commands** node β and everything β dominates if α and β are sisters.

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- Can you name the c-command relations in the tree?

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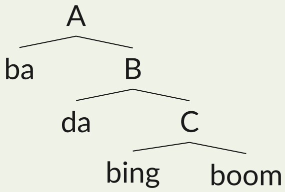

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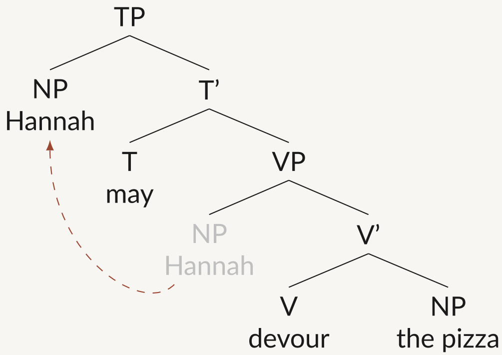

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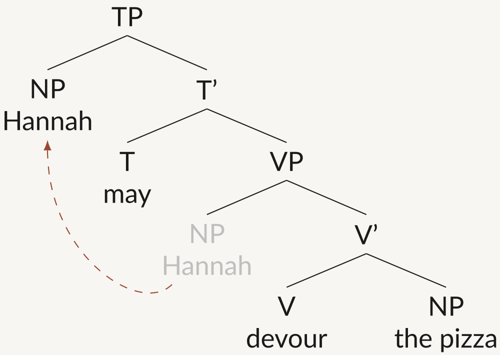

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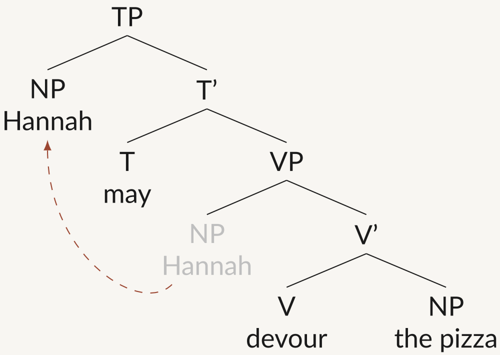

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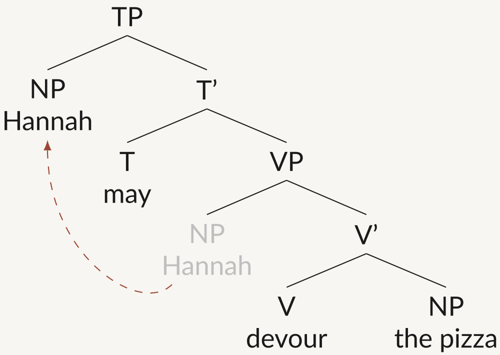

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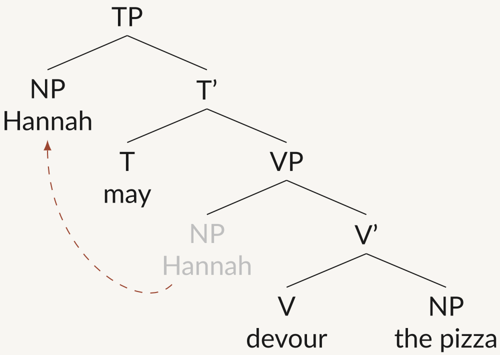

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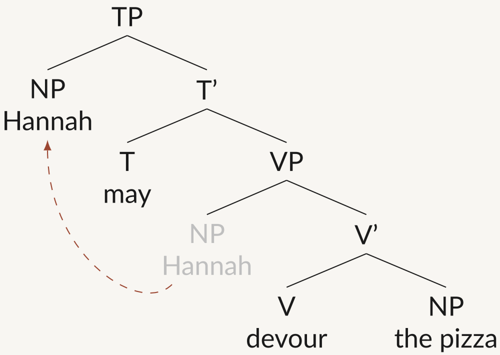

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- The specifier of V (SpecVP, which the NP *Hannah* originates in) c-commands V' and everything dominated by it.
- V' c-commands SpecVP.
- V and its complement (the NP *the pizza*) c-command each other.

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- Sisterhood is actually just the most local version of c-command!
- As such, the **hierarchical relations** that we have seen syntax be sensitive to so far can be **conflated into c-command**.

$\Rightarrow$ Selection just happens to be operate on the most local version of c-command.

A NOTE ON THE SUBJECT'S POSITION

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- The **closest suitable constituent** moves to the T head to fill the subject position.
- Only if there is no suitable constituent, like with zero-place predicates, is an expletive spelled out.

(18) a. Hannah laughed.
     b. It rained.
     c. *It Hannah laughed.

---

[6] This is likely one of the trickiest parts of basic syntactic theory, not because the empirical facts are controversial, but how to best capture them in the formal model is. I simplify *a lot* here.

A NOTE ON THE SUBJECT'S POSITION

Movement is the most economical solution to satisfy the EPP feature on (English) T:

- C-selectional features are checked locally, but the EPP feature triggers movement.

⇒ If there is a suitable constituent present in the structure, it will move to SpecTP.

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Movement is the most economical solution to satisfy the EPP feature on (English) T:

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$\Rightarrow$ If there is a suitable constituent present in the structure, it will move to SpecTP.

In languages with null subjects, free word order and postverbal subjects:

- We have **much fewer reasons to assume subject movement to SpecTP**.
- **T is projected nonetheless**: recall its primary purpose for tense and inflection.

Next week, you will learn more about cross-linguistic phenomena involving T.

INTERIM SUMMARY

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What about interrogatives? And embedded sentences?

The model we have so far can only generate simple declarative sentences.

EMBEDDED CLAUSES

(19) a. I think **that** it will rain.
b. I wonder **whether** it will rain.
c. I think **that** Hannah will devour the pizza.
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The elements in bold are called complementizers.

- Where do these complementizers stand relative to the other elements of the sentence?
- What does this imply for the structure we have so far?

WHAT AWAITS US TOMORROW

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- They **belong to the embedded clause** (rather than the main clause):

(20) a. I wonder **whether** she will sleep.
b. **Whether** she will sleep, I wonder.
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Complementizers are at the very top of the embedded clause!

⇒ Our tree will yet again have to grow.

Ask away!

APPENDIX: PSEUDOCLEFTING

Pseudoclefting is another constituency test:

- It is similar to clefting, which we saw in session 1, except that it uses a wh-word matching the constituent's category and features, and places the constituent we are testing at the very end rather than the beginning of the sentence.
- Applied to our example from session 1:

(21) *Sentence (a), clefting (b) and pseudoclefting (c)*

a. The mouse that stole the cheese was hiding from the cat **in the cabinet**.
b. It was **in the cabinet** that the mouse that stole the cheese was hiding from the cat.
c. **Where** the mouse that stole the cheese was hiding from the cat was **in the cabinet**.

APPENDIX: THETA ROLES ASSIGNED BY N AND P

(22) *Verbs (a, c) vs. deverbal nouns (b,d)*

a.	Brecht adapted Villon.	Brecht = agent; Villon = theme
b.	Brecht's adaptation of Villon	Brecht = agent; Villon = theme
c.	They destroyed the city.	they = agent; city = patient
d.	the destruction of the city	city = theme

(23) *Prepositions*

a.	He went to the store.	store = goal
b.	The mouse is hiding in the cabinet.	cabinet = location
c.	She hit the man with a stick.	stick = instrument